AMRITA VISHWA VIDYAPEETHAM
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MOFChat: Explainable, Data-Grounded MOF Recommendation with Validated LLM Planning, Multi-Objective Optimisation and Deep-Learning Surrogates

22AIE452 - Data Driven Material Modeling and Simulation | Final Review

Team_14

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Introduction

Metal-Organic Frameworks (MOFs) are crystalline porous materials made by linking metal ions with organic molecules, creating microscopic cages. Their tunable pores make them candidates for **carbon capture, hydrogen and methane storage, gas separation and catalysis**.

Selecting the right MOF is difficult: the CoRE-MOF database alone has **12,020 MOFs**, and a good choice has to balance several properties (surface area, pore size, density, metal sites) and needs domain knowledge.

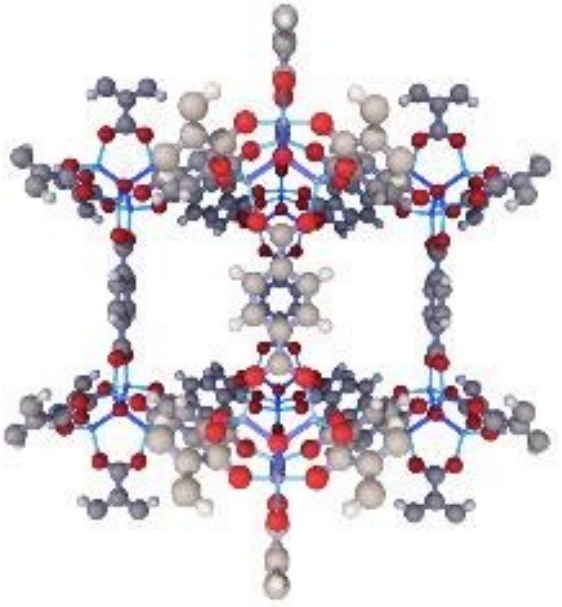
Problem statement

Existing LLM agents such as ChatMOF make MOF search conversational, but they generate code that can fail, depend on paid models, run into token limits and return numbers that are never checked against the data.

Aim

A conversational assistant that answers MOF questions exactly from the full database, ranks trade-offs between properties, predicts properties with uncertainty and explanations, runs on free or offline models, and verifies its own answers.

Terms: MOF = metal-organic framework. LLM = large language model (AI that reads and writes text, e.g. GPT-4, Gemini). CoRE MOF = Computation-Ready, Experimental MOF database. Domain knowledge = chemistry facts such as which pore sizes let which gases through.



MOF framework: metal nodes joined by organic linkers

Dataset at a glance	Value
MOFs in CoRE MOF 2019	12,020
Different metals	65
MOFs with zero accessible surface area	27 %

Base Paper: ChatMOF (Kang and Kim, 2024)

What the base paper did

- GPT-4 agent that plans tool calls (ReAct)
- Search: the LLM writes pandas code over the CoRE-MOF table
- Prediction: MOFTransformer (one fine-tuned model per property)
- Generation: genetic algorithm with PORMAKE

Task	Reported accuracy
Search	96.9 %
Prediction	95.7 %
Generation	87.5 %

https://www.nature.com/articles/s41467-024-48998-4

Drawbacks reported in the base paper

L1 No intrinsic materials knowledge (the LLM only plans tools)

L2 Token-context limits: 4 % of search tasks failed

L3 Reasoning and logic errors in generated code

L4 Restricted search space (about 100 parent structures)

L5 Narrow topology coverage (9 topologies)

L6 One pretrained model needed per property

L7 General, non-fine-tuned LLM (GPT-4)

L8 High compute cost (GA and simulations)

L9 Results are not validated against the data

Terms: ReAct = an agent loop where the LLM reasons, calls a tool, reads the result and repeats. pandas = Python library for tables. MOFTransformer = pretrained Transformer that predicts MOF properties from the 3-D structure. GA = genetic algorithm (evolves candidates by mutation and selection). PORMAKE = tool that builds MOF structures from building blocks. Topology = the connection pattern of metal nodes and linkers. Token limit = maximum amount of text an LLM can read at once.

Objectives and Our Novelty

Objectives

1. Exact answers over all 12,020 MOFs using free or local LLMs

2. Safe LLM planning with validation, self-correction and fallback

3. Multi-objective ranking with NSGA-II

4. Deep-learning and ML surrogate models with uncertainty and SHAP

5. Knowledge grounding by retrieval (RAG)

6. Measured results: benchmark, held-out set, classification and regression metrics

Base-paper limitation	MOFChat novelty
No materials knowledge	RAG over curated notes and the ChatMOF paper
Token limits	SQL filters first; the LLM sees at most 10 verified rows
Logic errors in code	The LLM writes a JSON plan; a validator checks and repairs it
Small search space	All 12,020 MOFs searched and ranked (NSGA-II)
Only 9 topologies	Every CoRE-MOF structure, 65 metals
One model per property	Our own XGBoost and FT-Transformer surrogates
Proprietary GPT-4	14 free or local LLMs with automatic fallback
High computation	Millisecond surrogate models instead of simulations
No validation	Answer validator, conformal intervals, SHAP and a benchmark

Terms: NSGA-II = Non-dominated Sorting Genetic Algorithm II (ranks options with several goals). SHAP = SHapley Additive exPlanations (how much each input pushed a prediction). RAG = retrieval-augmented generation (look up documents, then answer). SQL = database query language. JSON plan = structured text that says what to search. Surrogate = fast ML model that replaces a slow calculation. Held-out set = test questions never used while building.

Literature Review (2024-2026), part 1 of 3

#	Paper	Authors	Year, venue	Method	Contribution	Limitation	Use in MOFChat
1	**ChatMOF: an AI system for predicting and generating MOFs using LLMs**	Kang, Kim	2024, Nat. Commun.	GPT-4 agent writes pandas code; MOFTransformer; genetic algorithm	96.9 / 95.7 / 87.5 % on search / prediction / generation	Token limits, code errors, answers not checked	Base paper; we use validated JSON plans and SQL
2	**Augmenting LLMs with chemistry tools (ChemCrow)**	Bran et al.	2024, Nat. Mach. Intell.	GPT-4 with 18 expert tools in a reason-act loop	Planned real syntheses; found a new chromophore	Needs GPT-4; tool outputs not cross-checked	Tool calling; our tools are deterministic and tested
3	**MOFDiff: coarse-grained diffusion for MOF design**	Fu, Xie, Rosen et al.	2024, ICLR	Equivariant diffusion over building blocks	Valid, novel MOFs for CO2 capture	Needs simulation to validate	Generation is future work
4	**TableRAG: million-token table understanding**	Chen et al.	2024, NeurIPS	Schema and cell retrieval before the LLM	Table QA on very large tables	Numeric filters still judged by the LLM	We filter with SQL first
5	**Chain-of-Table: evolving tables in the reasoning chain**	Wang et al.	2024, ICLR	LLM plans table operations step by step	Best results on WikiTQ and TabFact	No domain validation of operations	We validate every planned operation
6	**Detecting hallucinations in LLMs using semantic entropy**	Farquhar, Kossen, Kuhn, Gal	2024, Nature	Entropy over the meanings of sampled answers	Detects confabulations without task data	Misses consistent errors; needs many samples	We check answers against the data instead

Terms: QA = question answering. WikiTQ / TabFact = public table question-answering benchmarks. Confabulation = a confident but made-up LLM answer. Equivariant diffusion = generative model that respects rotations of a structure. CO2 capture = trapping carbon dioxide from gas mixtures.

Literature Review (2024-2026), part 2 of 3

#	Paper	Authors	Year, venue	Method	Contribution	Limitation	Use in MOFChat
7	**Harnessing LLMs to collect and analyze MOF property data (L2M3)**	Lee, Kang, Bae, Kim	2025, JACS	LLM text mining of over 40,000 articles	Experimental MOF database and synthesis recommender	Extraction errors from text	Source of real labels for future uptake models
8	**MOF-ChemUnity: literature-informed LLMs for MOF research**	Pruyn et al.	2025, JACS	Multi-agent LLM links literature names to CSD structures	Literature-grounded answers via a knowledge graph	Large infrastructure	Our RAG grounds answers in notes and papers
9	**Accurate predictions on small data with a tabular foundation model (TabPFN)**	Hollmann et al.	2025, Nature	Transformer pretrained on synthetic tables	Beats tuned GBDT on small tables	About 10k rows / 500 features at most	Possible surrogate for our 8k-row tables
10	**TabM: tabular deep learning with parameter-efficient ensembling**	Gorishniy, Kotelnikov, Babenko	2025, ICLR	MLP with BatchEnsemble	Beats attention-based tabular models	Generic benchmarks only	Matches our finding: DL close to trees
11	**LLaMP: LLM for high-fidelity materials knowledge retrieval**	Chiang et al.	2025, EMNLP	Hierarchical ReAct agents over Materials Project	Grounded answers with a confidence metric	Needs the Materials Project API	Same grounding idea with a local MOF database
12	**System of agentic AI for the discovery of MOFs (MOFGen)**	Inizan et al.	2025, arXiv	LLM proposes compositions; diffusion and quantum-chemistry agents	Synthesizable MOF candidates	Heavy compute; preprint	We verify answers against the database

Terms: JACS = Journal of the American Chemical Society. CSD = Cambridge Structural Database of crystal structures. Knowledge graph = facts stored as linked entities. GBDT = gradient-boosted decision trees (e.g. XGBoost). BatchEnsemble = cheap way to train many networks that share weights. Materials Project = public database of inorganic materials.

Literature Review (2024-2026), part 3 of 3

#	Paper	Authors	Year, venue	Method	Contribution	Limitation	Use in MOFChat
13	**L2M3OF: a large language multimodal model for MOFs**	Cui et al.	2025, arXiv	Crystal encoder projected into an LLM	Better than text-only LLMs on properties	Needs structure files and fine-tuning	Multimodal input is future work
14	**Conversational LLM agent for synthesis of MOF catalysts (MOFsyn Agent)**	Lin et al.	2025, ACS Nano	RAG and ML agent guiding synthesis	Faster Ni@UiO-66(Ce) catalyst optimisation	Single reaction system	RAG and ML combined, as in MOFChat
15	**The rise of generative AI for MOF design and synthesis**	Duan et al.	2025, arXiv (perspective)	Review of VAEs, diffusion models and LLM agents	Roadmap to closed-loop discovery	Feasibility and data gaps	MOFChat sits at the screening stage
16	**LitMOF: LLM multi-agent for literature-validated MOF database correction**	Kim, Kim, Kim	2025, arXiv	Agents check CSD structures against papers	189,567 structures; 9,227 entries repaired	Preprint	We also audit the data (ETL checks)
17	**Enhancing spatial reasoning in LLMs for MOF structure prediction (MOF-LLM)**	Pan et al.	2026, arXiv	Qwen-3 8B with spatial pretraining, SFT and RL	35.78 % match rate, 0.04 s per structure	Low match rate	Small open LLMs can be specialised
18	**An LLM agent for end-to-end computational materials discovery (MAESTRO)**	Chen et al.	2026, arXiv	Agent mines papers, builds a database, screens MOFs	Finds wet-flue-gas separation candidates	Preprint	MOFChat is a verified query layer

Terms: Multimodal = uses more than one kind of input (crystal structure and text). VAE = variational autoencoder (generative model). SFT = supervised fine-tuning. RL = reinforcement learning. Preprint = paper not yet peer-reviewed (arXiv). Closed-loop discovery = AI proposes, lab tests, results feed back.

Research Gap from the Literature

Gap	Evidence in 2024-2026 work	What MOFChat does
Answers are rarely verified against data	ChatMOF, ChemCrow and LLaMP trust the LLM's final text; semantic entropy needs many samples.	A deterministic answer validator checks IDs, numbers and comparison claims against the database.
Dependence on proprietary LLMs	Most agents use GPT-4.	14 free or local models with fallback; works fully offline with Llama 3.
Few benchmarks for query planning	Accuracy is reported on the authors' own question sets.	64-question development set, 20 held-out questions, model comparison and ablation.
Tabular DL vs trees on MOF descriptors	TabPFN, TabM and Grinsztajn et al. compare on generic data.	FT-Transformer, MLP, XGBoost and GP compared on MOF regression and classification.

Terms: Deterministic = always gives the same output for the same input (normal code, not an LLM). Proprietary = paid, closed model. Ablation = removing one part to measure what it adds. Tabular DL = deep learning on tables. MLP = multilayer perceptron (simple neural network). GP = Gaussian process.

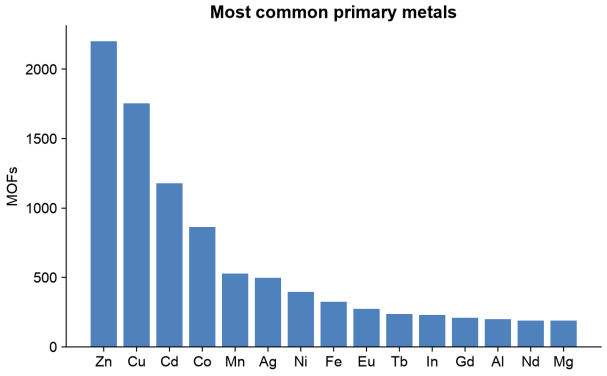
Dataset: CoRE MOF 2019 and Preprocessing

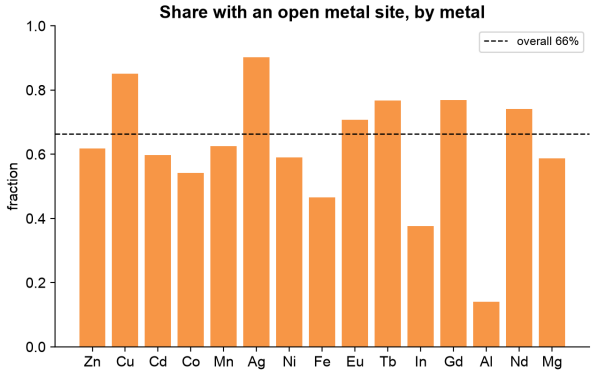
Category	Attributes
Identification	MOF_ID (CSD refcode)
Pore geometry	LCD, PLD, LFPD (Å)
Density	Density (g/cm³)
Surface area	Accessible (m²/g, m²/cm³), non-accessible
Porosity	Void fraction, pore volume, non-accessible volume
Chemistry	Metal type, open metal site, OMS type

12,020 experimental MOFs and 17 columns, computed with Zeo++ using a 3.31 Å (N2) probe.

Preprocessing (ETL) findings

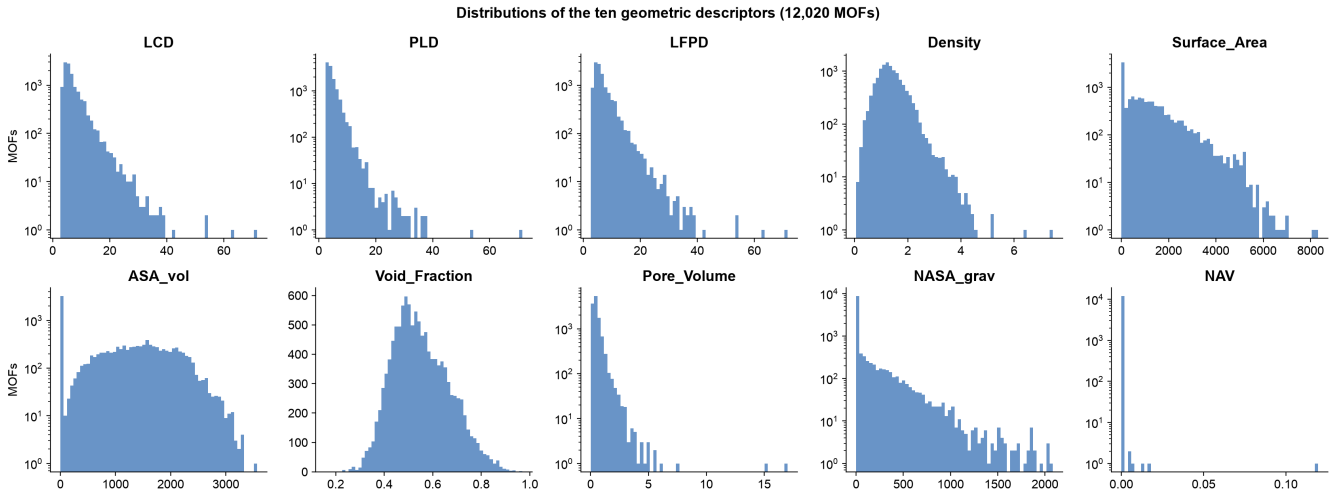
Check	Finding and action
Missing values, duplicates	None
Two column pairs identical	Duplicates dropped
Pore volume = void fraction / density	Derived column; documented
Zero surface area (3,252 MOFs)	All have PLD < 3.31 Å; physics rule
Density header said cm³/g	Values are g/cm³; label fixed
2,087 multi-metal MOFs	Separate mof_metals table
Feature scaling	log1p and z-score

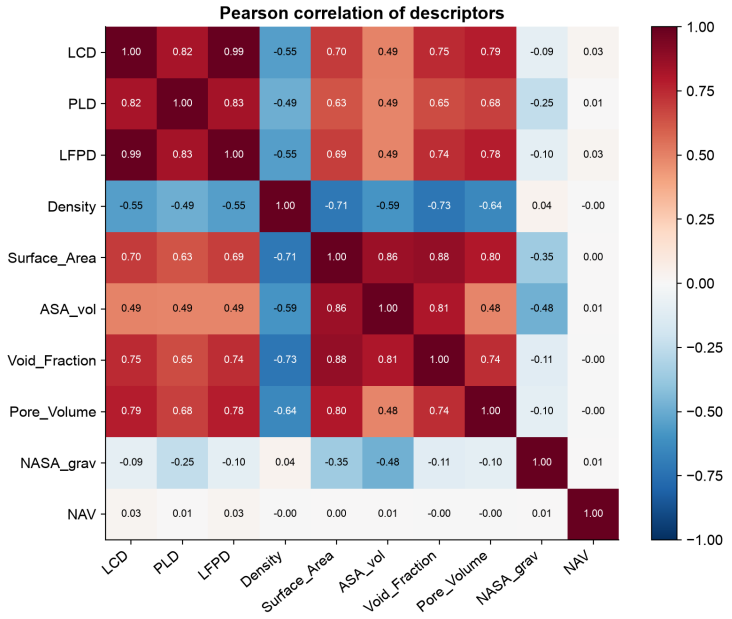




Terms: LCD = largest cavity diameter (biggest sphere that fits in a cage). PLD = pore limiting diameter (narrowest window a molecule must pass). LFPD = largest free pore diameter (biggest sphere that can travel through). OMS = open metal site. CSD refcode = MOF ID in the Cambridge Structural Database. Zeo++ = software that computes pore geometry. ETL = extract, transform, load. log1p = log(1 + x). z-score = (value - mean) / standard deviation.

Feature Extraction: Descriptor Distributions and Correlations





Pearson correlation of the ten descriptors

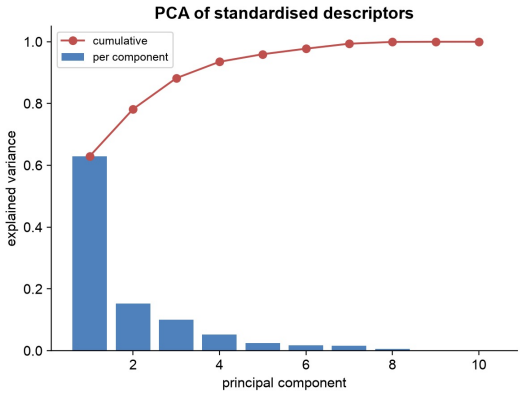
Observations

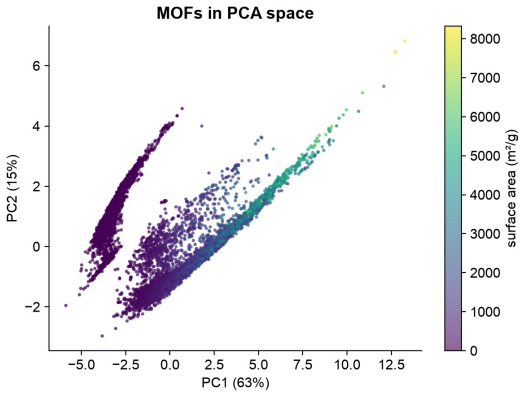
- Most descriptors are right-skewed, so they are log-transformed and standardised before the FT-Transformer
- 27 % of MOFs have exactly zero accessible surface area (the spike at 0)
- Strongest correlations: LCD and LFPD (+0.99), Surface_Area and Void_Fraction (+0.88), Surface_Area and ASA_vol (+0.86), PLD and LFPD (+0.83)
- Pore volume is derived from void fraction and density, so it is left out when predicting void fraction

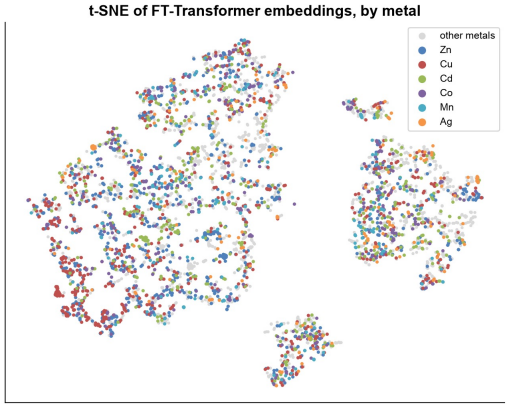
Terms

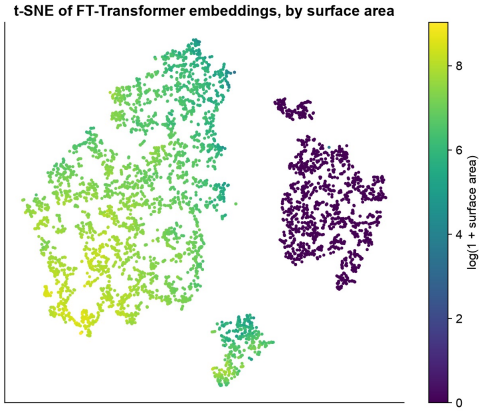
- **LCD:** largest cavity diameter
- **PLD:** pore limiting diameter (narrowest window)
- **LFPD:** largest free pore diameter along a path
- **ASA / ASA_vol:** accessible surface area per gram / per cm³
- **Void fraction:** share of the volume that is empty
- **Pore volume:** empty volume per gram
- **NASA / NAV:** non-accessible area / volume
- **Pearson r:** -1 to +1, how linearly two values move together

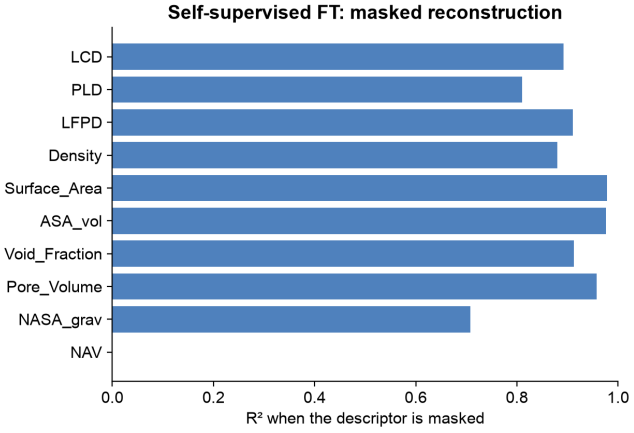
Feature Extraction: PCA, Learned Embeddings and Reconstruction

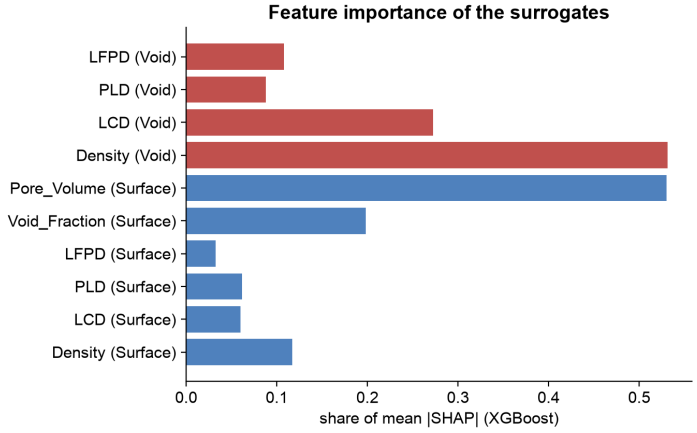








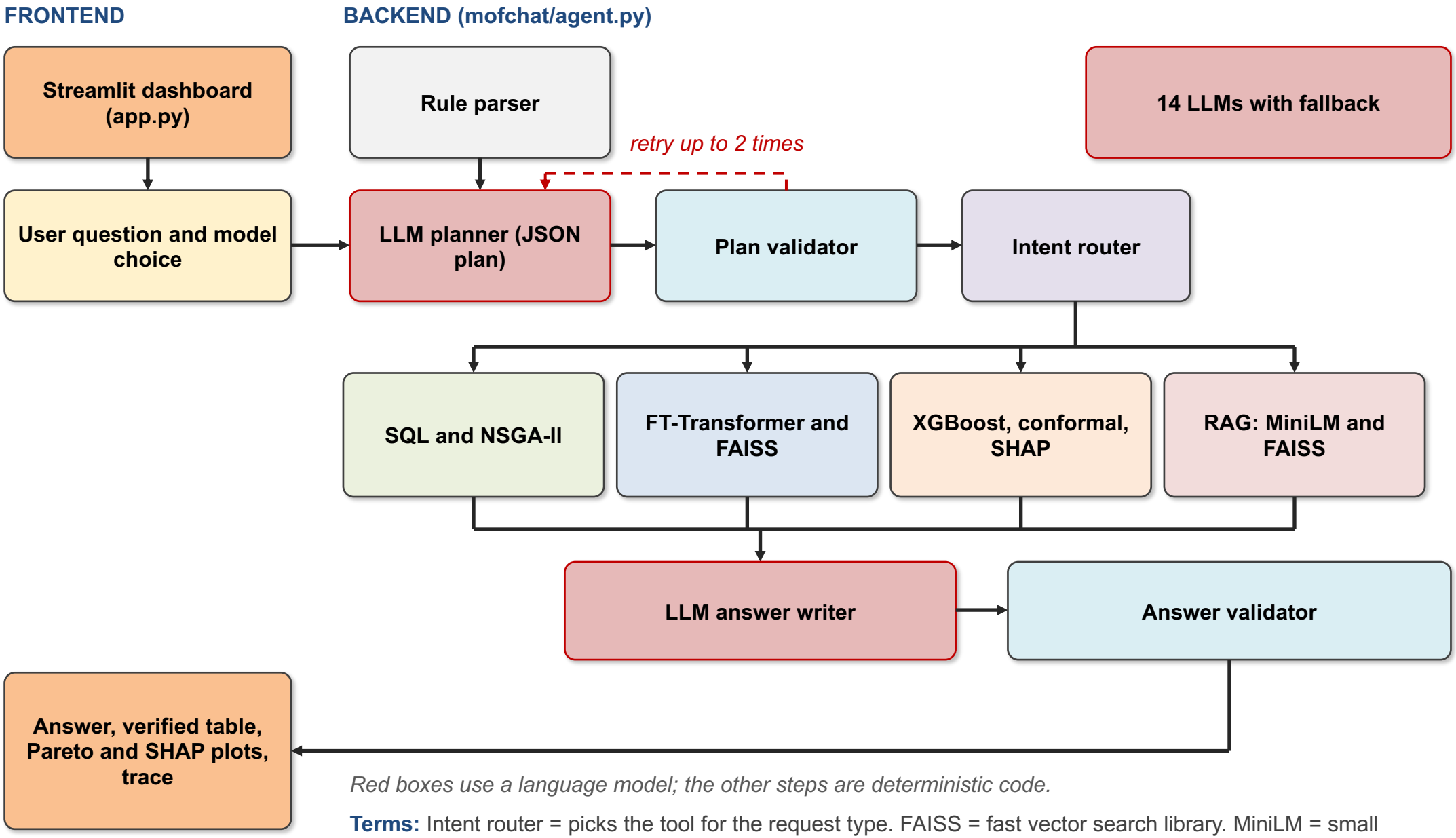




- **PCA:** the first two components explain 78 % of the variance
- **t-SNE of the 64-d FT-Transformer embeddings:** smooth surface-area gradient; metals form local clusters
- **Masked reconstruction:** R^2 0.978 for surface area and 0.913 for void fraction when hidden
- **SHAP:** pore volume and void fraction drive surface area; density drives void fraction

Terms: PCA = principal component analysis (finds the main directions of variation). t-SNE = method that draws high-dimensional data as a 2-D map. Embedding = a vector of numbers that represents a MOF. Masked reconstruction = hide some inputs and let the model rebuild them. R^2 = share of variance explained (1 = perfect).

Overall Pipeline: Dashboard to Backend



Red boxes use a language model; the other steps are deterministic code.

Terms: Intent router = picks the tool for the request type. FAISS = fast vector search library. MiniLM = small sentence-embedding model. Conformal = 90 % prediction interval. Fallback = next model used when one fails.

Pipeline Architecture: Offline Training and Online Inference

Offline (run once)



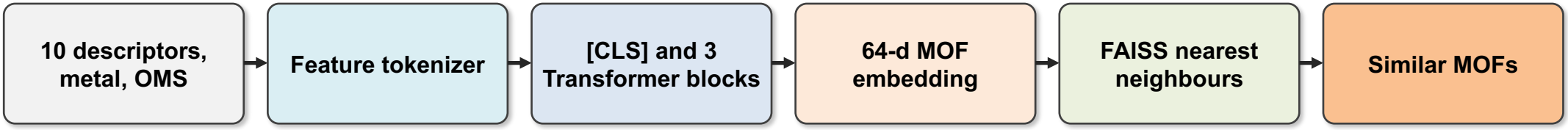
• Dataset: 12k CoRE-MOFs • ETL: Clean & SQLite • FT-Transformer: 64-d SSL • Surrogates: XGBoost • Index: S-BERT + FAISS RAG

Scripts: setup_db.py, train_ft.py, train_property_models.py, setup_vectorstore.py, evaluate.py, evaluate_dl.py

Online (per question)



Feature extraction inside the FT-Transformer



In review 1, 384-d MOF embeddings were matched to a Sentence-BERT question vector. The two were never trained together, so this path was replaced by plan, validate and SQL (exact answers rose from 9.4 % to 100 %).

Terms: SSL = self-supervised learning (learns from the data itself, no labels). S-BERT = Sentence-BERT. Regex = text-pattern rules. Levenshtein = number of letter edits between two words. [CLS] = extra summary token. L2-normalised = scaled to length 1. Cosine = similarity of two vectors (1 = same direction).

Models Used (1): Language and Embedding Models

Model	Type	How it works	Role in MOFChat
Gemini 3.1 Flash-Lite	LLM, cloud (free)	Transformer decoder that predicts the next token; we prompt it with the schema, rules and 9 examples at temperature 0 in JSON mode	Default planner and answer writer
GPT-OSS 120B / 20B, Qwen 3.8 27B (Groq)	LLMs, cloud (free)	Same principle; open-weight models served by Groq	Fallback planners and writers
Nemotron 3, Gemma 4 (OpenRouter)	LLMs, cloud (free)	Open models through an OpenAI-compatible API	Further fallbacks
Llama 3 / 3.1 8B, Qwen 2.5 7B (Ollama)	LLMs, local	Small open models running on the laptop CPU	Offline planning and answers
all-MiniLM-L6-v2 (Sentence-BERT)	Embedding model, 384-d	6-layer Transformer; mean-pooled sentence vectors compared by cosine similarity	Embeds questions and knowledge chunks for RAG
FT-Transformer autoencoder	Embedding model, 64-d	Each descriptor becomes a token; 30 % are masked and the [CLS] state reconstructs all ten	MOF embeddings for similarity search
FAISS	Vector index	Inner product on normalised vectors equals cosine similarity; fast exact search	Nearest-neighbour search for RAG and similarity

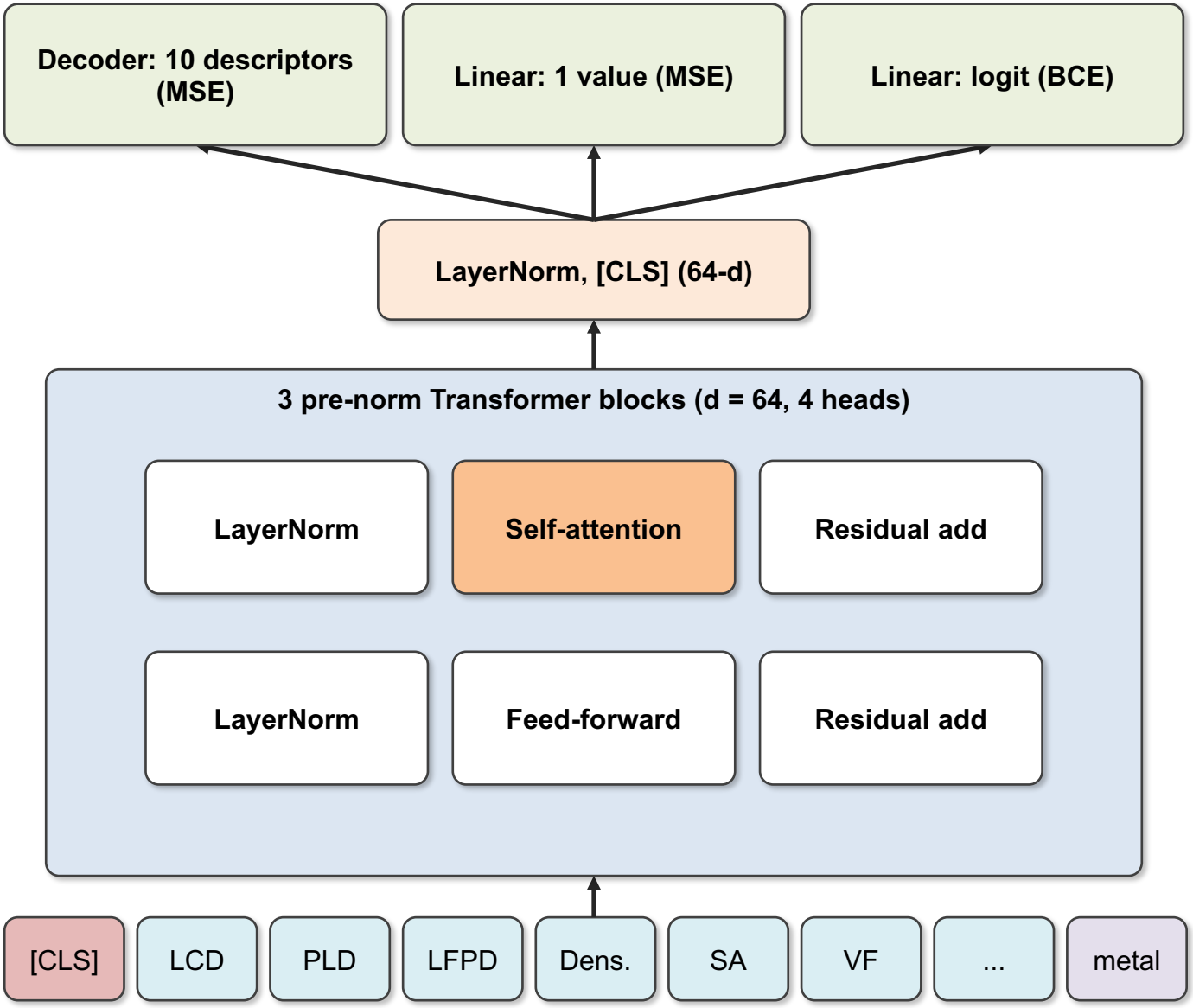
Terms: Token = a piece of a word or one input feature. Temperature 0 = always choose the most likely output (repeatable). JSON mode = output forced to structured JSON. Open-weight = model weights are public. 384-d / 64-d = a vector of 384 / 64 numbers. Mean-pooled = average of the word vectors.

Models Used (2): Prediction Models and Algorithms

Model	Type	How it works	Role in MOFChat
XGBoost	Gradient-boosted trees	Hundreds of small decision trees added one after another, each fitting the errors of the previous ones	Main surrogate model; OMS classifier
FT-Transformer regressor / classifier	Deep learning	Same Transformer backbone with a linear head; MSE or cross-entropy loss; early stopping on validation loss	Surface area, void fraction; OMS classification
MLP classifier	Deep learning	Two hidden layers of 128 ReLU units with a metal embedding; cross-entropy loss	Deep-learning baseline for OMS
Gaussian process	Bayesian regression	RBF kernel with one length scale per feature plus noise; gives a mean and an uncertainty	Regression baseline with 95 % intervals
Ridge, logistic regression	Linear models	Linear models with an L2 penalty	Baselines
NSGA-II	Multi-objective algorithm	Sorts candidates into Pareto fronts, then prefers well-spread points (crowding distance)	Ranks MOFs on several objectives
SHAP (TreeSHAP)	Explainability	Shapley values split each prediction into per-feature contributions	Explains every prediction
Split-conformal prediction	Uncertainty	90th percentile of absolute errors on a calibration split gives a ± interval	90 % prediction intervals

Terms: MSE = mean squared error. Cross-entropy (BCE) = loss for yes/no classification. ReLU = max(0, x). RBF kernel = similarity that decays with distance. L2 penalty = keeps weights small. Pareto front = set of best trade-offs. Calibration split = data kept aside only to size the interval.

Model Architecture: FT-Transformer (Gorishniy et al., 2021)



Input tokens: numeric value times a learned vector plus bias, categorical embeddings and a learned [CLS] token

Configuration

Item	Value
Parameters (embedding)	114,378
Parameters (classifier)	104,449
Blocks / heads / d	3 / 4 / 64
Optimiser	AdamW, lr 0.001
Batch size	256
Early stopping	validation loss
SSL masking	30 % of tokens
SSL training	38 epochs
Regressor (surface area)	163 epochs
Classifier (OMS)	best epoch 31

Terms: d = numbers per token. Heads = parallel attention units. Self-attention = each token weighs every other token. LayerNorm = rescales each token. Residual add = adds the input back. FFN = feed-forward layer. AdamW = optimiser. lr = learning rate. Epoch = one pass over the training data.

LLM Planning: Plan, Validate, Execute

"Which zinc MOFs have high surface area and low density?"

```
{ "intent": "search",
  "metals": ["Zn"],
  "objectives": [
    {"column": "Surface_Area", "direction": "max"},
    {"column": "Density", "direction": "min"} ],
  "limit": 5 }
```

Executor: whitelisted columns, bound parameters, read-only SQL on zinc MOFs, then NSGA-II ranking for the two objectives

Intents: search, count, lookup, similar, predict, knowledge, out of scope

Validator checks

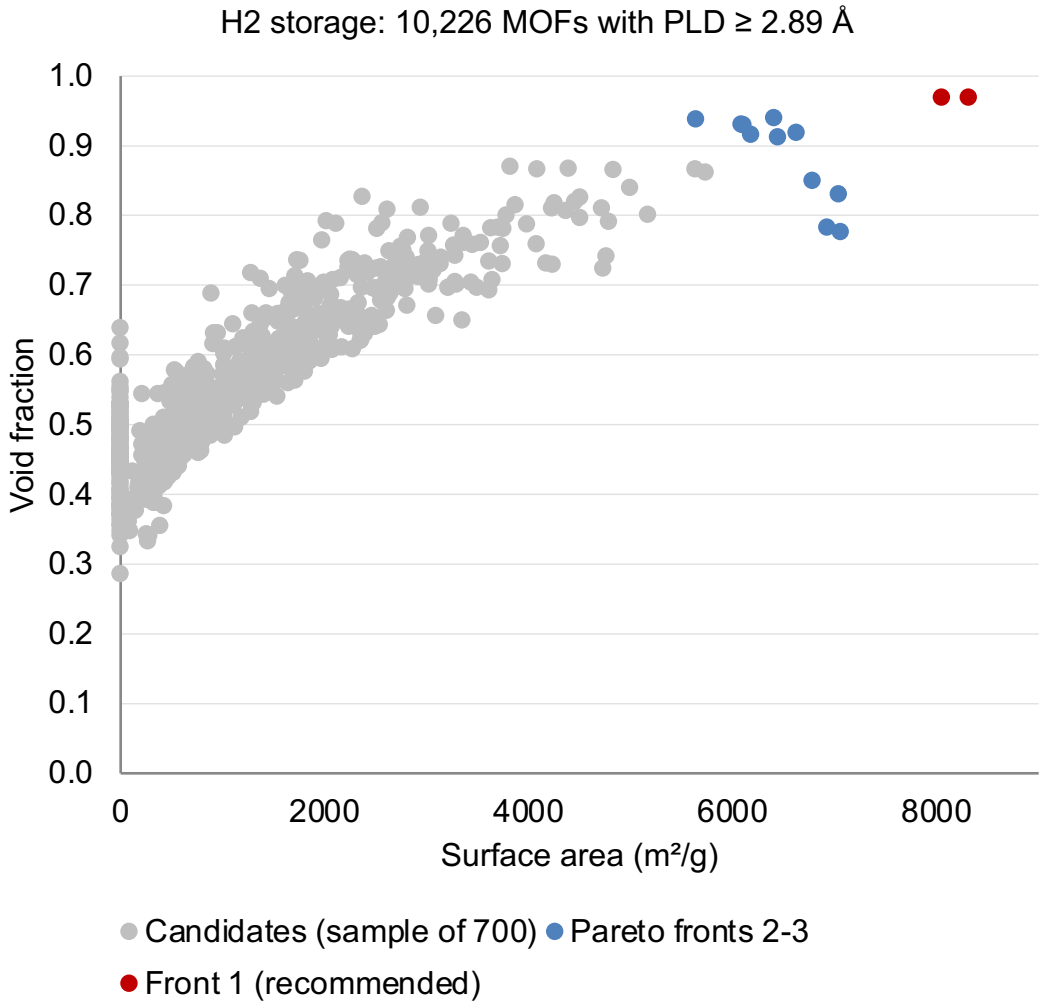
- Column aliases resolved from one registry
- Units, operators and percentages normalised
- Impossible or invented values rejected
- Thresholds not in the question become preferences
- MOF IDs not in the question dropped
- Superlatives written as constraints repaired
- Application presets expanded (H2, CH4, CO2, sieving, catalysis)
- Objectives not asked for in an application question dropped

If errors remain after two corrections, the rule-parser plan is used. If no LLM is reachable, the rules and a template answer are used.

Terms: Intent = type of request. Whitelisted columns = only known column names allowed. Bound parameters = numbers passed separately, so the query cannot be changed. Read-only = cannot modify data. Superlative = 'highest', 'lowest'. Preset = ready-made rules for an application. Rule parser = regex-based question reader.

Multi-Objective Ranking with NSGA-II

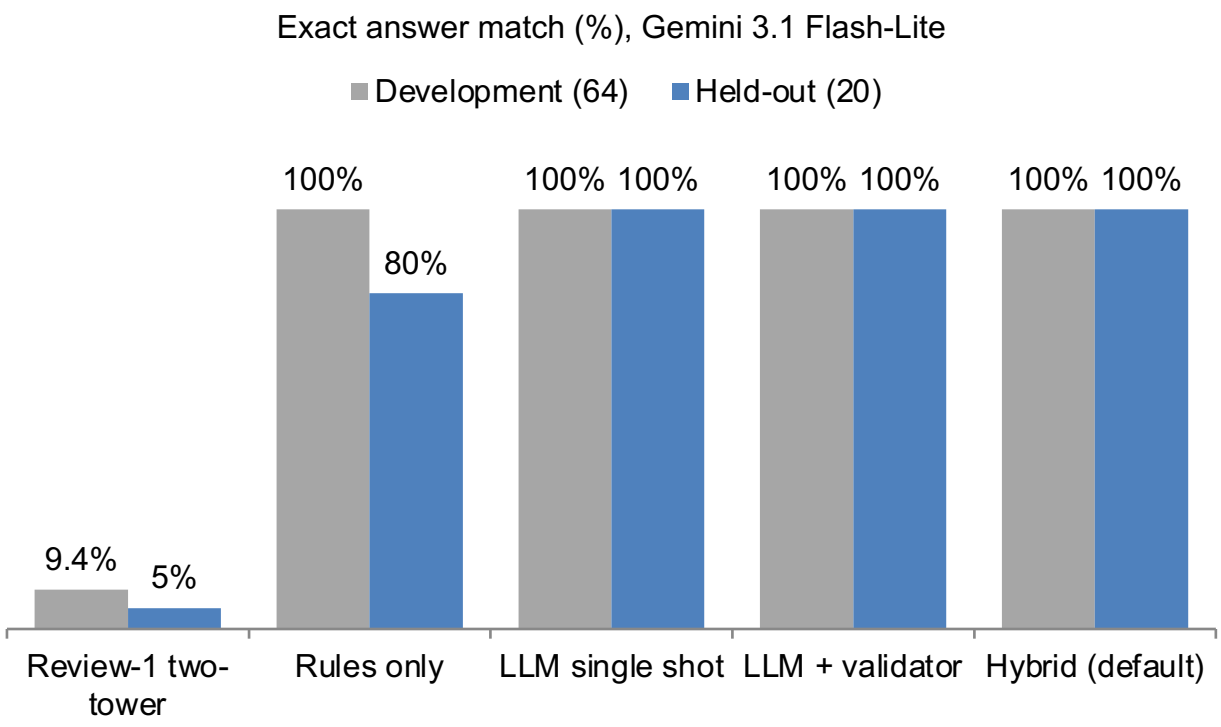
- **Domination:** A beats B if A is at least as good on every objective and better on one
- **Non-dominated sorting:** front 1 holds the MOFs no other MOF beats; remove them and repeat
- **Crowding distance:** prefers MOFs in sparse regions, giving diverse trade-offs
- **Two objectives:** exact O(n log n) sweep; the worst 25 % are screened out first
- **Speed:** all 12,020 MOFs in 0.02 s (2 objectives) and 2.6 s (3)

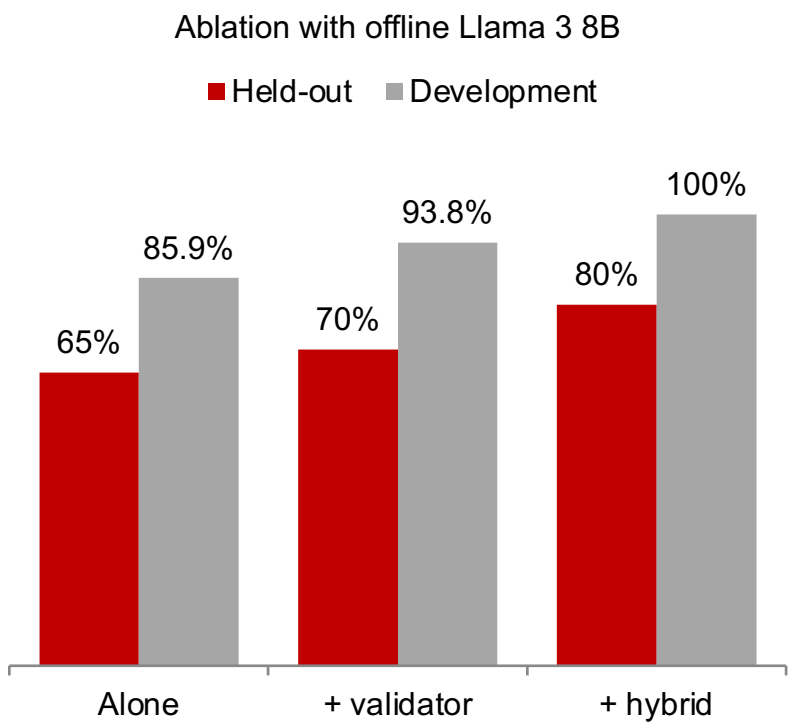


Application	Heuristic rule
H2 storage	PLD ≥ 2.89 Å; max surface area and void fraction
CH4 storage	PLD ≥ 3.80 Å; max volumetric area and void fraction
CO2 capture	Open metal site; PLD ≥ 3.30 Å; min LCD
Separation A/B	Kinetic diameter of A ≤ PLD ≤ that of B
Catalysis	Open metal site; PLD ≥ 5 Å

Terms: Pareto front = MOFs not beaten on all objectives at once. Objective = a property to maximise or minimise. O(n log n) = time grows only a little faster than the number of MOFs. Kinetic diameter = effective size of a gas molecule (H2 2.89 Å, CO2 3.3 Å, CH4 3.8 Å).

Results: Retrieval Benchmark





- An answer counts as correct only if it returns exactly the MOFs of a hand-written gold plan. Constraint precision@k on the held-out set: review 1 41.8 %, hybrid 100 %.
- The rules were tuned on the development set, so the held-out questions (written later, never used for tuning) give the fair estimate.
- The validator and hybrid planner add 15 points for the offline model and cost less than 0.01 s; for the cloud model they change nothing.
- ChatMOF reports 96.9 % on its own search questions; the question sets differ.

Terms: Gold plan = the correct plan written by hand. Precision@k = share of returned MOFs that really satisfy the question. Development set = 64 questions used while building. Held-out set = 20 later questions never used for tuning. Single shot = LLM alone, no corrections. Two-tower = review-1 method matching question and MOF vectors.

Results: Free LLM Comparison and Answer Validation

Model	Held-out alone	Held-out hybrid	Dev hybrid	Median call	Provider errors
Gemini 3.1 Flash-Lite (default)	100 %	100 %	100 %	1.5 s	13 / 84
GPT-OSS 120B (Groq)	95 %	95 %	100 %	2.4-4.7 s	0
Qwen 3.8 27B (Groq)	95 %	95 %	100 %	20 s	0
Nemotron 3 Super 120B	95 %	95 %	-	2.0 s	0
Llama 3 8B (local)	65 %	80 %	100 %	1.5 s	0
Gemini 3.5 Flash	-	-	-	-	64 / 84 (quota)

Answer validation (16 full answers per model)

Model	Drafts with a problem	Rewritten	Still flagged	s per answer
Gemini 3.1 Flash-Lite	0 of 16	0	0	9.9
Llama 3 8B (local)	2 of 16	1	0	9.3
GPT-OSS 120B	2 of 16	2	1	30.2
Qwen 3.8 27B	0 of 16	0	0	26.4

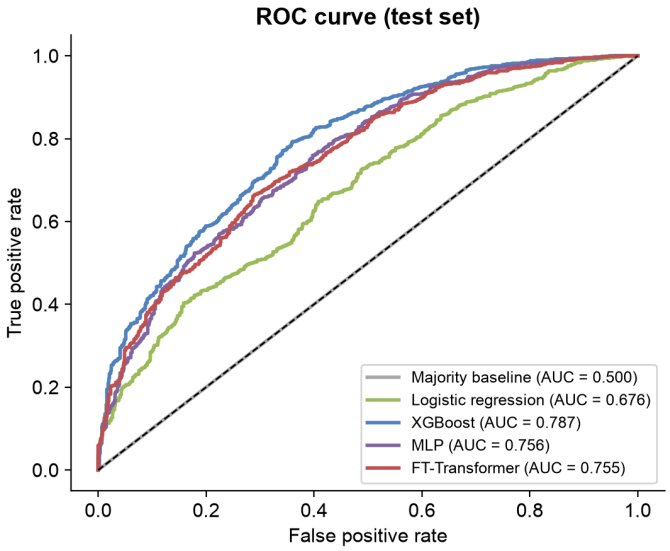
Terms: Hybrid = LLM plan plus facts the rule parser is sure of. Median call = typical time for one planning request. Provider error = the free API refused or failed the call. Quota = daily request limit. Draft = the LLM's first answer before checking.

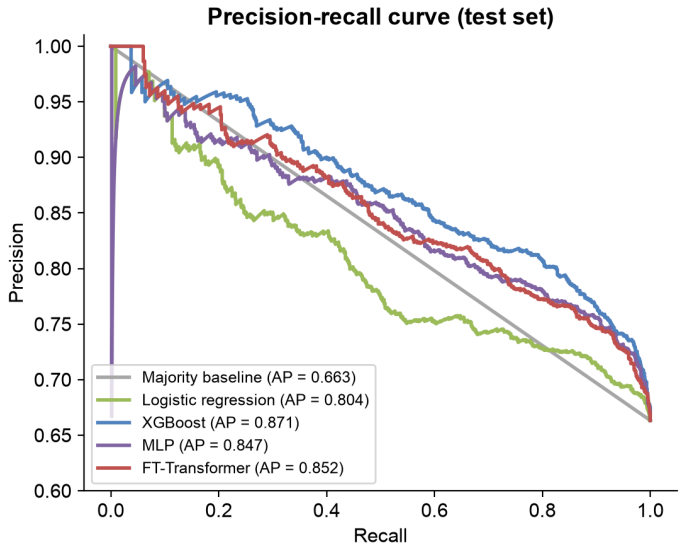
Examples caught by the validator

- Llama 3 invented two densities and a "90 %"; the ID OFEUN06 was corrected to OFERUN06
- GPT-OSS 120B claimed "Pareto front 1" for MOFs ranked on front 2
- Llama 3 named MOFs that do not exist ("MOF-210") in a prediction answer
- GPT-OSS cited a prompt header "[WHY]" as a source; it is now removed

Results: Open-Metal-Site Classification (DL vs ML)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Majority baseline	0.663	0.663	1.000	0.798	0.500	0.663
Logistic regression	0.693	0.708	0.916	0.798	0.676	0.804
XGBoost	0.750	0.765	0.901	0.827	0.787	0.871
MLP	0.734	0.759	0.878	0.814	0.756	0.847
FT-Transformer	0.728	0.755	0.873	0.810	0.755	0.852



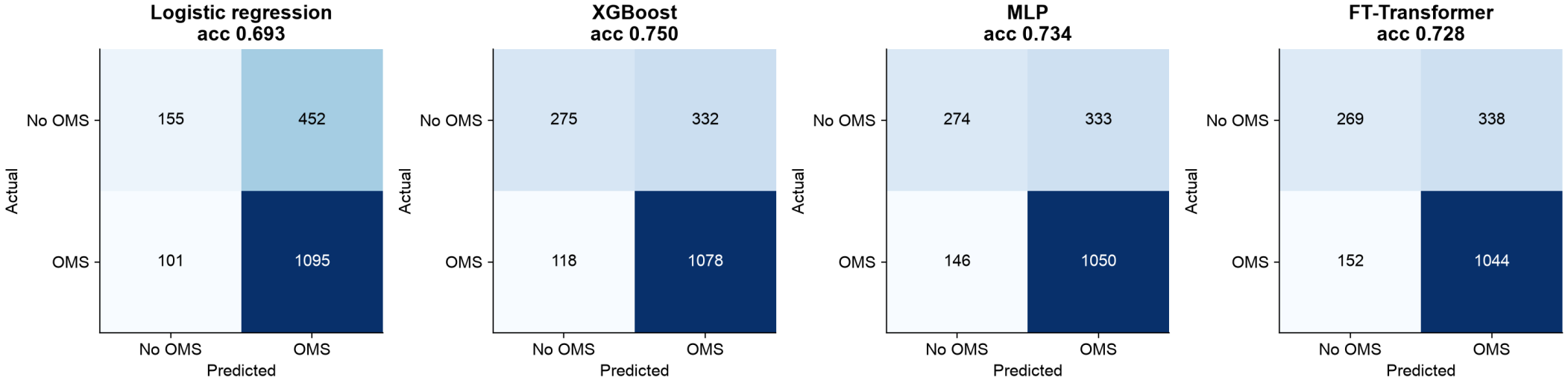


Task: predict whether a MOF has an open metal site from its 10 geometric descriptors and primary metal. 66.4 % of MOFs have one. OMS_Type is not used as an input.

Split: 8414 / 1803 / 1803 MOFs (train / validation / test, stratified = same OMS share in each)

Terms: Accuracy = share of correct predictions. Precision = of MOFs predicted OMS, share that are right. Recall = of real OMS MOFs, share found. F1 = balance of precision and recall. ROC-AUC = chance a real OMS MOF scores higher than a non-OMS one (0.5 = guessing). PR-AUC = area under the precision-recall curve. Majority baseline = always predict the most common class.

Results: Confusion Matrices of the OMS Classifiers

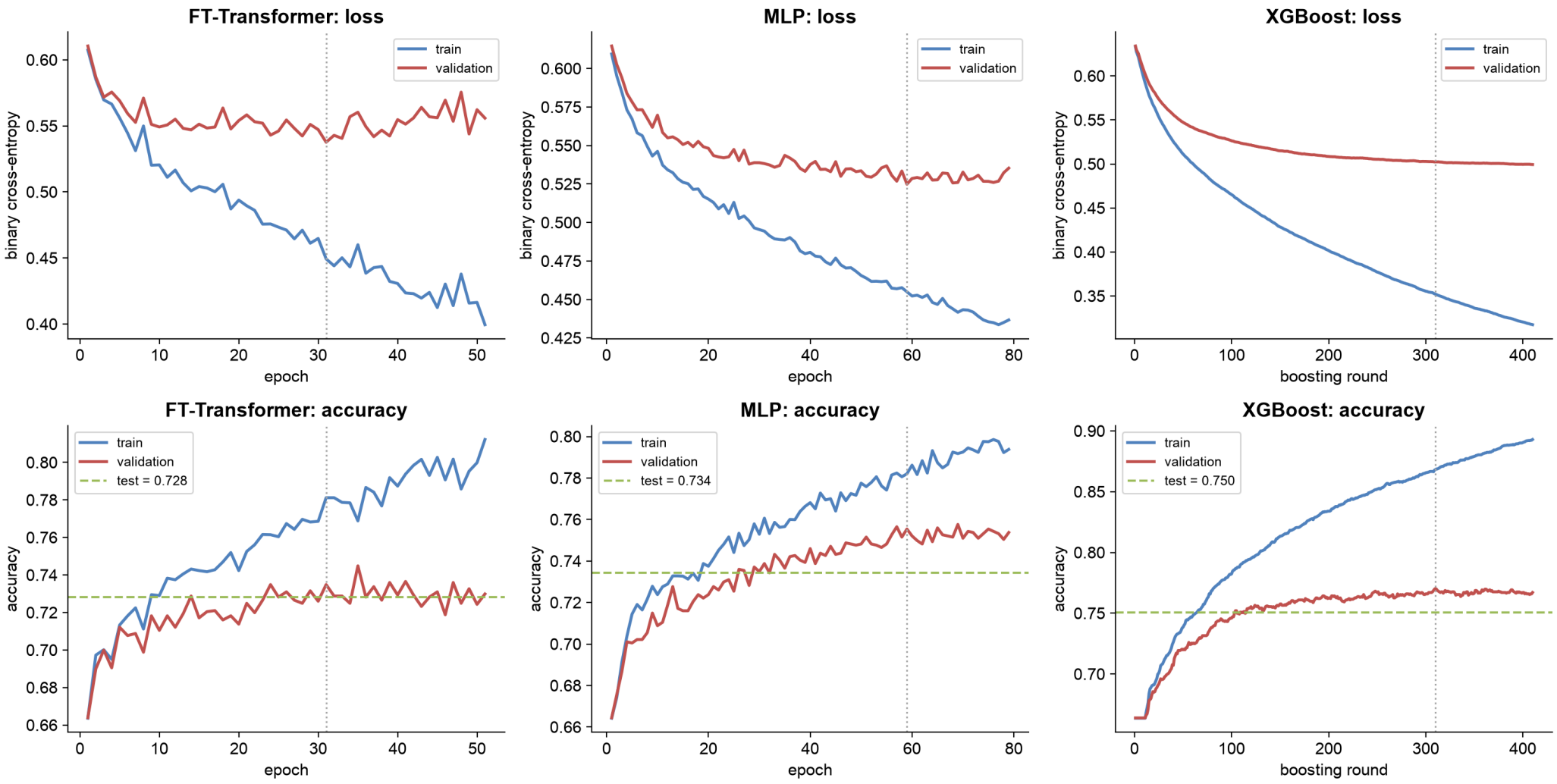


Model	True negative	False positive	False negative	True positive	Train / val / test accuracy
XGBoost	275	332	118	1078	0.868 / 0.770 / 0.750
MLP	274	333	146	1050	0.782 / 0.755 / 0.734
FT-Transformer	269	338	152	1044	0.781 / 0.735 / 0.728

- All trained classifiers beat the majority baseline (66.3 % accuracy, ROC-AUC 0.50). XGBoost is best and the two deep-learning models are close behind.
- Whether a metal site is open depends on coordination chemistry that pore geometry and metal type only partly describe, so accuracy levels off near 75 %.

Terms: Confusion matrix = table of actual vs predicted classes. True positive = has OMS, predicted OMS. False positive = predicted OMS but has none. False negative = has OMS, predicted none. True negative = no OMS, predicted none. Overfitting = learning the training data too closely (train accuracy much higher than test).

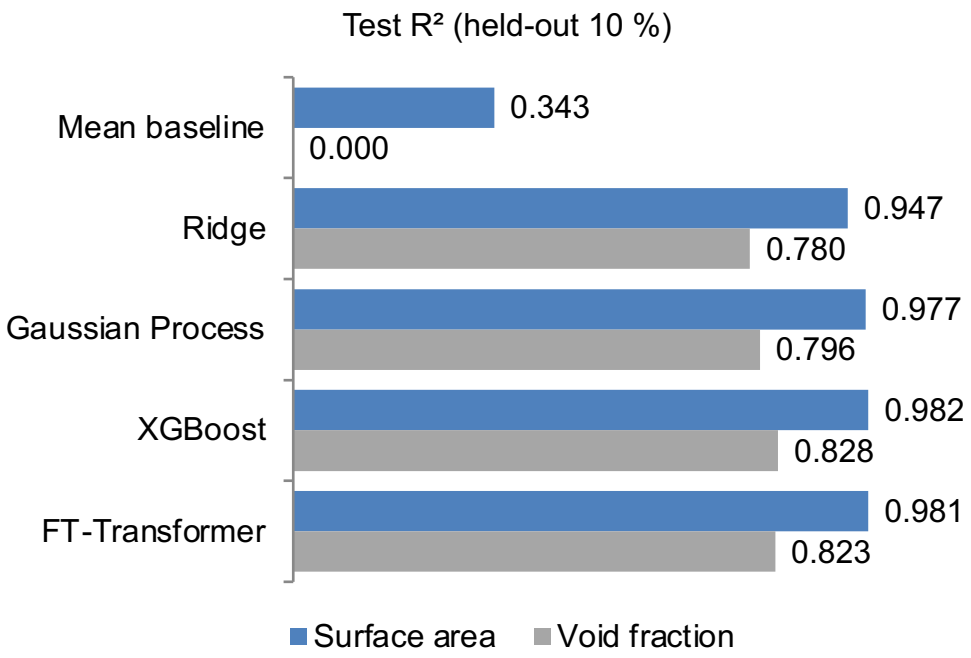
Results: Train, Validation and Test Accuracy and Loss

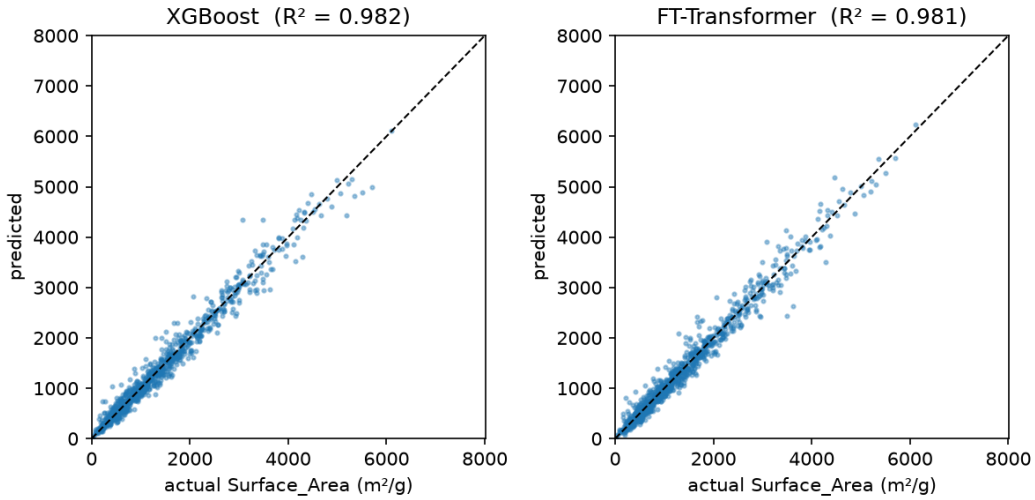


Blue: train, red: validation, green dashed: test, dotted: best epoch. FT-Transformer best epoch 31 of 51; MLP 59; XGBoost best round 310.

Terms: Loss = how wrong the model is (lower is better). Epoch = one pass over the training data. Boosting round = one tree added in XGBoost. Early stopping = keep the epoch with the lowest validation loss. Validation set = data used to choose that epoch; test set = untouched final check.

Results: Property Prediction (Regression)





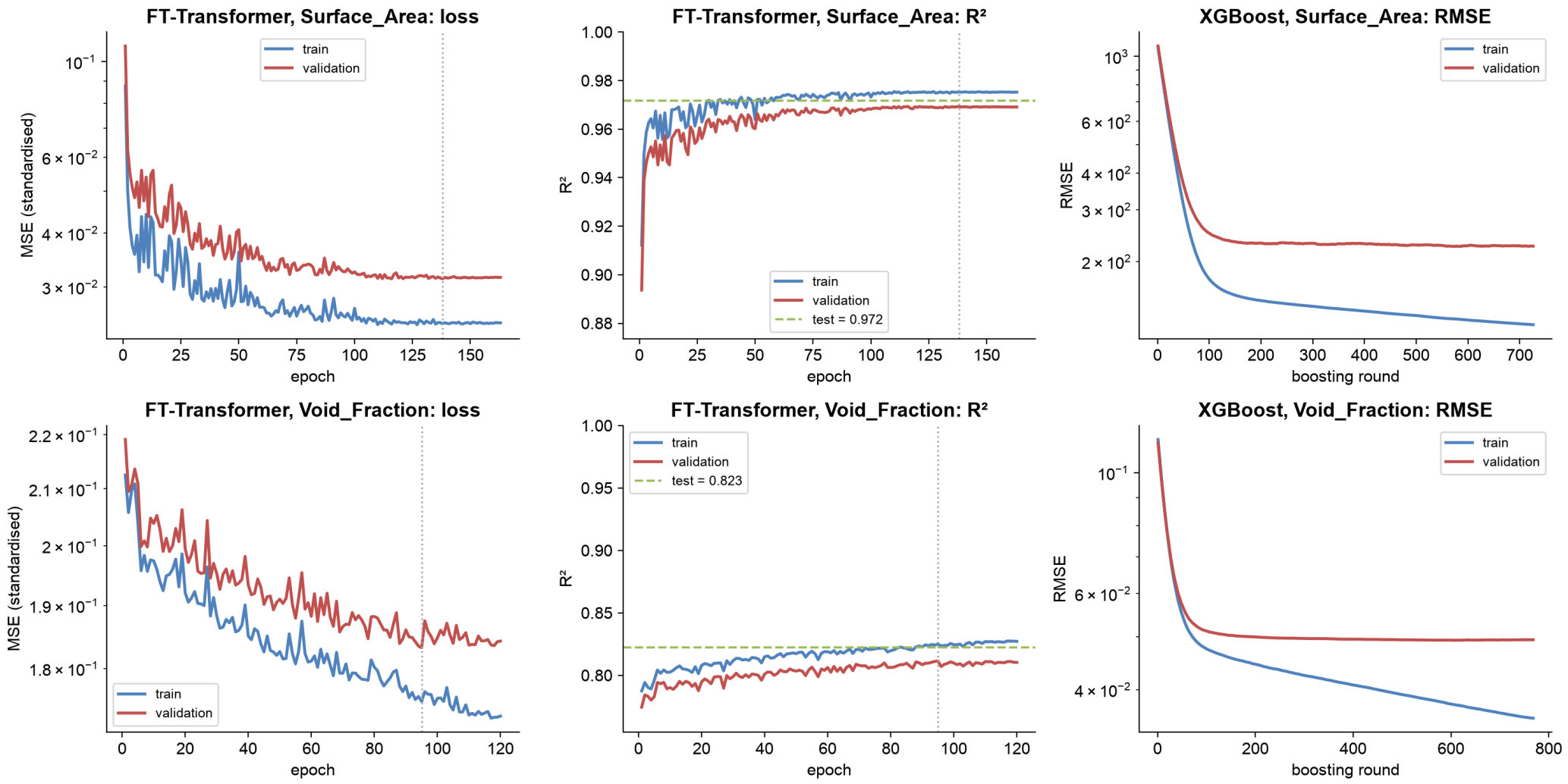
Predicted vs actual surface area on test MOFs

Target	XGBoost MAE	XGBoost R² train / val / test *	FT R² train / val / test *	90 % interval coverage	GP 95 % coverage
Surface area (m²/g)	90.0	0.987 / 0.959 / 0.972	0.975 / 0.969 / 0.972	90.5 %	93.9 %
Void fraction	0.037	0.899 / 0.820 / 0.828	0.825 / 0.811 / 0.823	88.4 %	95.1 %

* R² on MOFs with accessible pores; the physics rule handles the zeros exactly (100.0 % on test). XGBoost trains in 0.25 s, the FT-Transformer in 111.58 s.

Terms: Regression = predicting a number. R² = share of variance explained (1 = perfect). MAE = mean absolute error. Parity plot = predicted vs actual (perfect on the diagonal). Interval coverage = share of true values inside the 90 % interval. GP = Gaussian process.

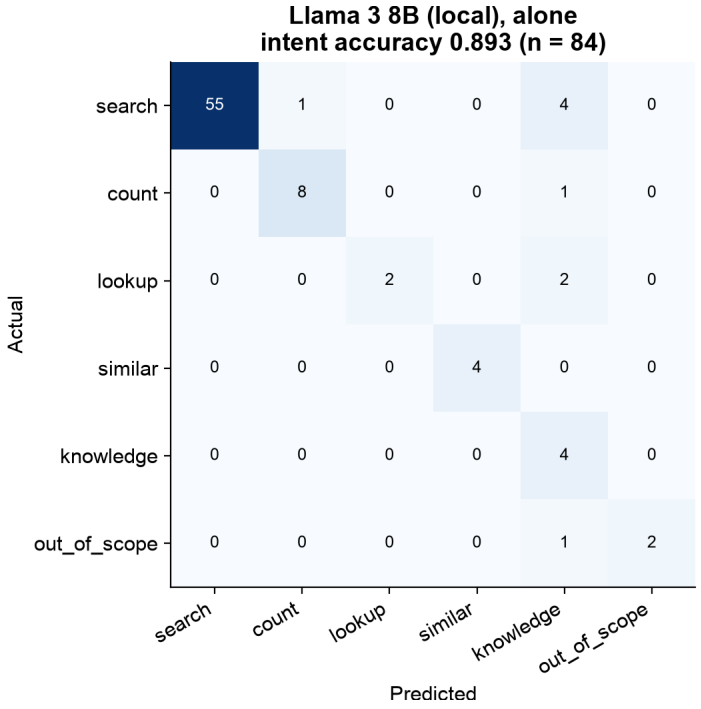
Results: Regression Learning Curves

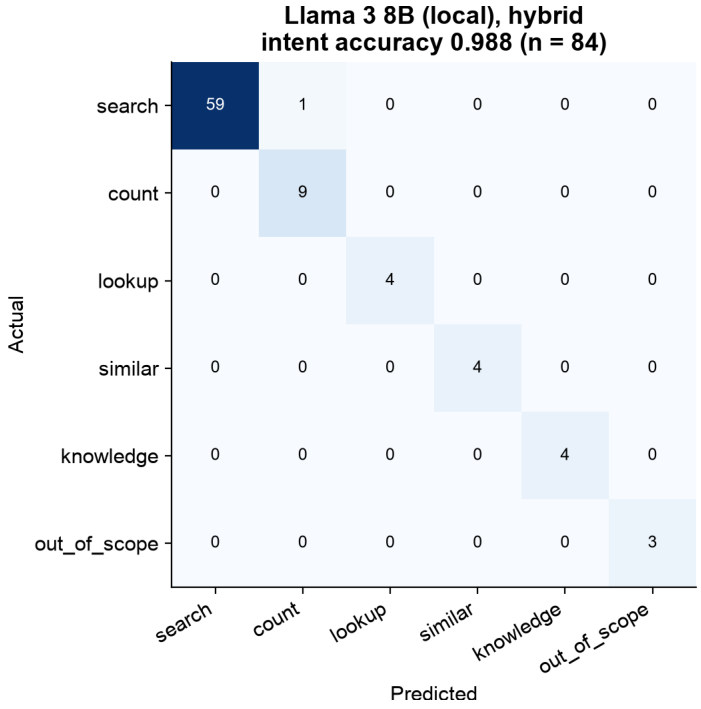


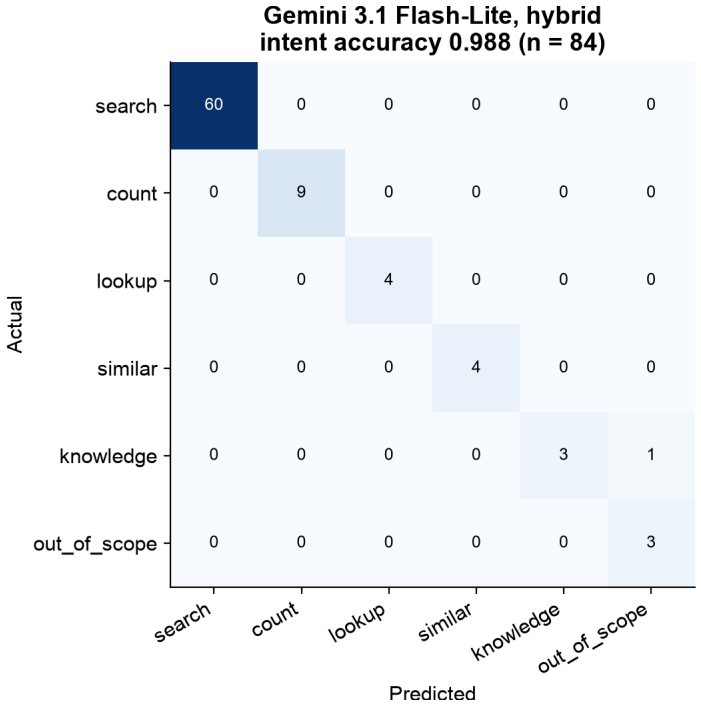
Blue: train, red: validation, green dashed: test R². FT-Transformer best epoch 138 (surface area) and 95 (void fraction); XGBoost best round 576 and 618.

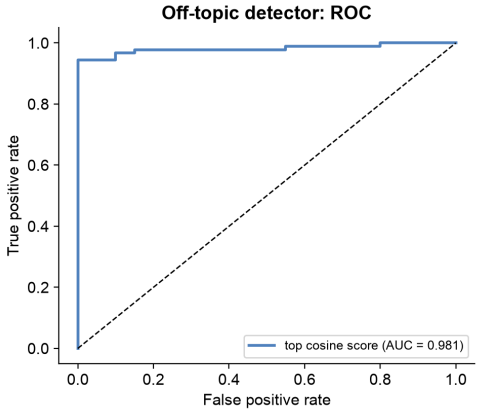
Terms: MSE (standardised) = squared error after scaling the target. RMSE = root mean squared error, in the target's units. Log scale = each grid step is ten times the last.

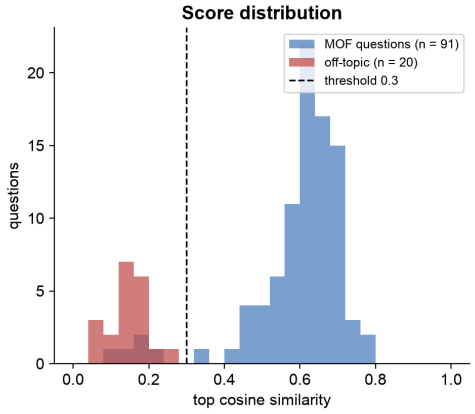
Results: Intent Classification and Off-Topic Detection

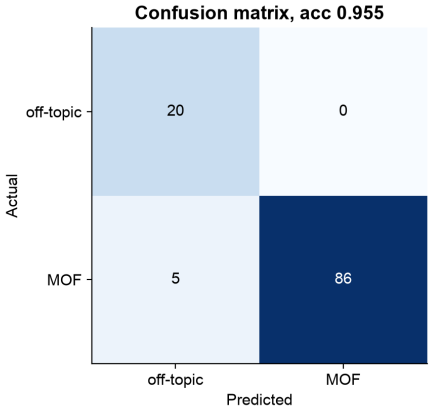










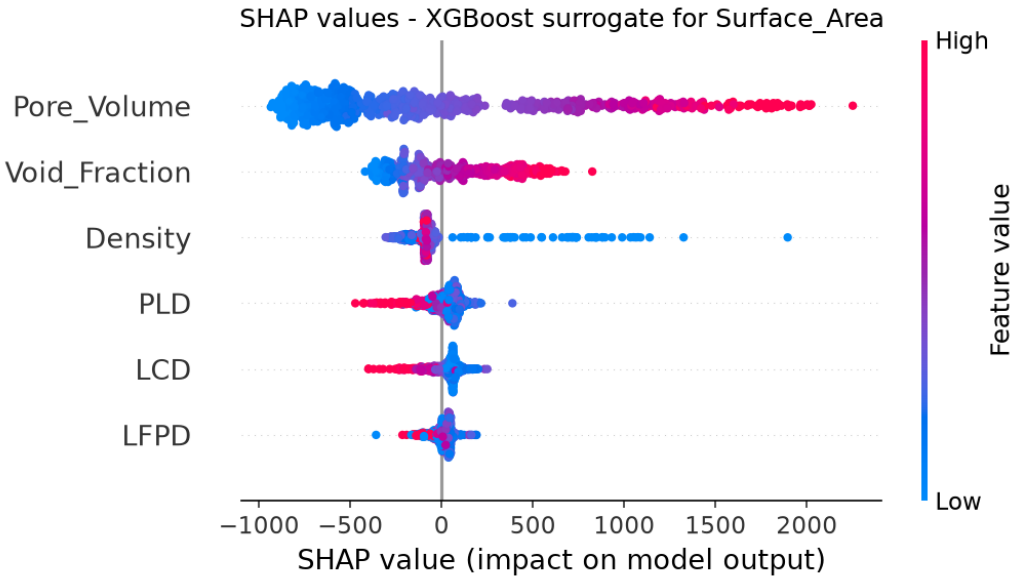


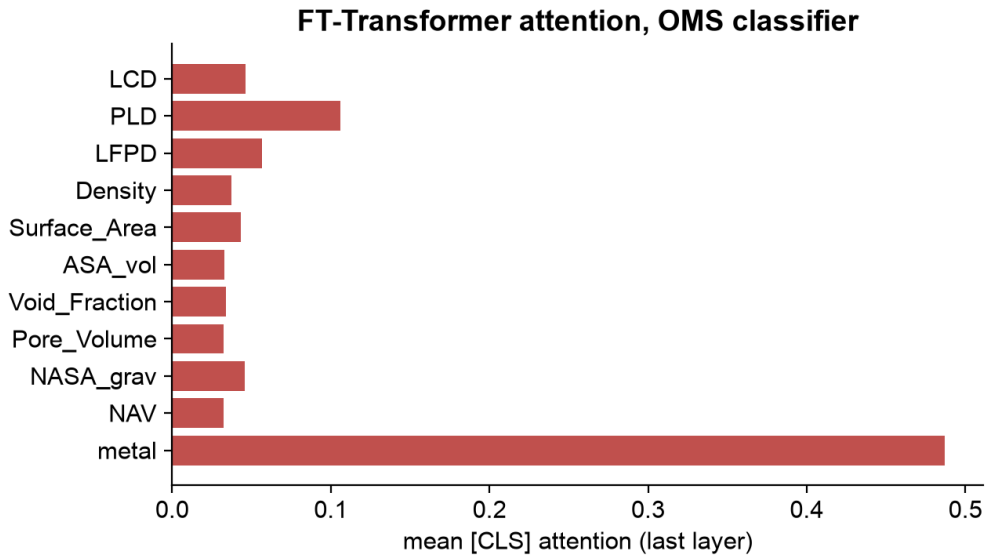
Planner	Accuracy	Macro F1
Rules planner	0.964	0.950
Llama 3 8B, alone	0.893	0.802
Llama 3 8B, hybrid	0.988	0.990
Flash-Lite, hybrid	0.988	0.952

Off-topic detector: ROC-AUC 0.981, accuracy 0.955 on 91 MOF questions and 20 off-topic questions (cosine threshold 0.3).

Terms: Intent = request type (search, count, ...). Macro F1 = F1 averaged equally over all intents. Off-topic = not about MOFs. Cosine threshold = minimum text similarity to count as a MOF question.

Results: Explainability and Similarity Search





Neighbours from (1,000 queries)	Descriptor gap (σ)	Same metal	Same OMS flag
FT-Transformer embedding	0.077	27.8 %	70.2 %
k-NN on raw descriptors	0.068	22.8 %	66.4 %
Random MOFs	0.952	8.0 %	55.7 %

- Pore volume and void fraction dominate surface-area predictions; density dominates void fraction
- Knowledge retrieval: hit@1 75 %, hit@3 83 %
- Every prediction in the application shows its SHAP contributions and a 90 % interval

Terms: SHAP value = how much a feature pushed one prediction up or down. Attention = how much the model looks at each input. σ = standard deviation. k-NN = k nearest neighbours on the raw numbers. hit@k = correct source among the top k results.

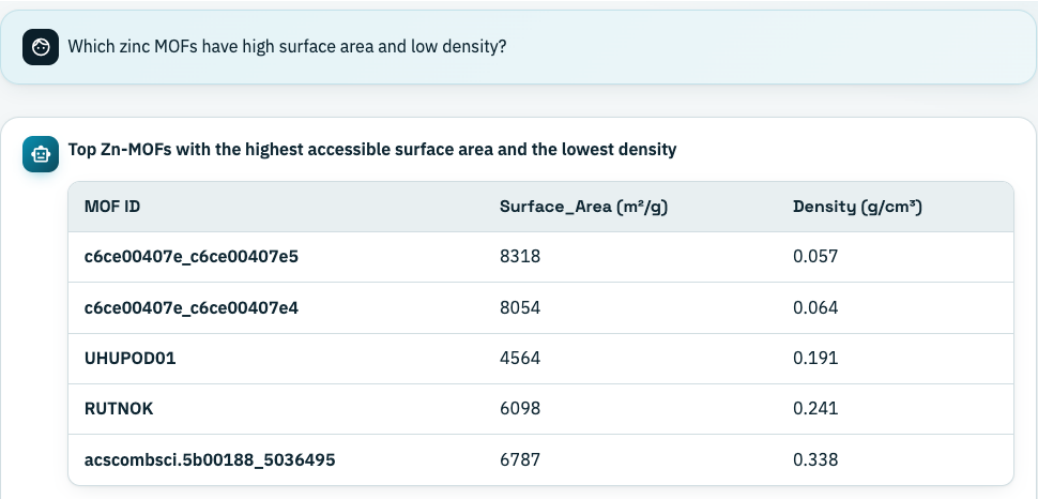
Results: ChatMOF Limitations Addressed

ChatMOF limitation	What MOFChat does	Status
No intrinsic knowledge	RAG over curated notes and the ChatMOF paper	**Done**
Token-context limits	SQL first; the LLM sees at most 10 rows	**Done**
Logic errors in LLM code	JSON plan, validator, whitelisted SQL	**Done**
Restricted search space	All 12,020 MOFs searched and ranked	**Done**
Narrow topology coverage	All CoRE-MOF structures, 65 metals	**Done ***
Pretrained model per property	Own surrogate models and OMS classifier trained on the table	**Done ***
General, non-fine-tuned LLM	Domain prompts, few-shot examples, presets	**Partly**
High compute cost	Millisecond screening; XGBoost trains in 0.25 s	**Done**
Unvalidated output	Answer validator, intervals, SHAP, benchmark	**Done**

* for recommending existing MOFs and for properties in the table; no structure generation and no gas uptake without uptake data

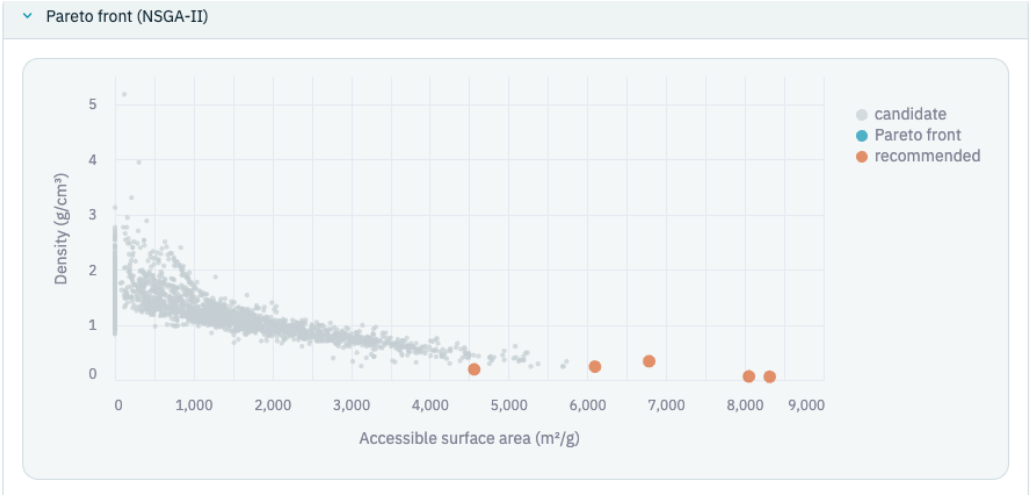
Terms: Few-shot = worked examples placed in the prompt. Fine-tuning = retraining an LLM on our own data. Gas uptake = how much gas a MOF adsorbs.

Application: Streamlit Dashboard



MOF ID	Surface_Area (m²/g)	Density (g/cm³)
c6ce00407e_c6ce00407e5	8318	0.057
c6ce00407e_c6ce00407e4	8054	0.064
UHUPOD01	4564	0.191
RUTNOK	6098	0.241
acscombsci.5b00188_5036495	6787	0.338

Verified result table for a zinc MOF question



NSGA-II Pareto plot for the same question

Page	What it shows
Chat	Answer, verified table, Pareto plot, SHAP, sources, plan, SQL, validator report, model used
Explore and optimise	Filters and two-objective NSGA-II with a Pareto plot
Predict and explain	Prediction, 90 % interval, SHAP bars, FT-Transformer attention
Evaluation	Benchmark, model comparison, parity and SHAP plots
About	Method summary and data sources

Sidebar: choice of language model (or none), planner mode (hybrid, LLM or rules), and the status of every component.

Terms: Streamlit = Python library for web dashboards. Parity plot = predicted vs actual values. Planner mode = how the question is read: hybrid (LLM + rules), LLM only, or rules only.

Conclusion and Future Work

Conclusion

- Exact answers over all 12,020 MOFs: 5 % (review 1) to 100 % on held-out questions with the default model, 80 % fully offline
- Validated LLM planning with 14 free or local models and automatic fallback
- NSGA-II multi-objective ranking with application presets
- Surface-area surrogate: R^2 0.982; the 90 % interval covers 90.5 % of test MOFs
- Open-metal-site classification: ROC-AUC 0.787 (XGBoost), 0.755 (FT-Transformer), 0.756 (MLP)
- An answer validator checks every MOF ID, number and comparison in the LLM's answer

Limitations and future work

Limitation	Future work
No gas-uptake labels	Add GCMC uptake data and reuse the surrogate pipeline
No structure generation	Diffusion or GA generator (e.g. MOFDiff)
Offline model at 80 %	LoRA fine-tuning of Llama 3 on plan examples
Free-text reasoning not checked	Sentence-level checks such as semantic entropy
Heuristic presets	Learned rules once uptake data exists
Descriptor-only embeddings	Structure or multimodal encoders (L2M3OF)

Terms: GCMC = grand canonical Monte Carlo (simulation of gas uptake). LoRA = low-rank adaptation (cheap fine-tuning of an LLM). GA = genetic algorithm. Semantic entropy = uncertainty measured over the meanings of several sampled answers. Multimodal encoder = model that reads crystal structures as well as text.

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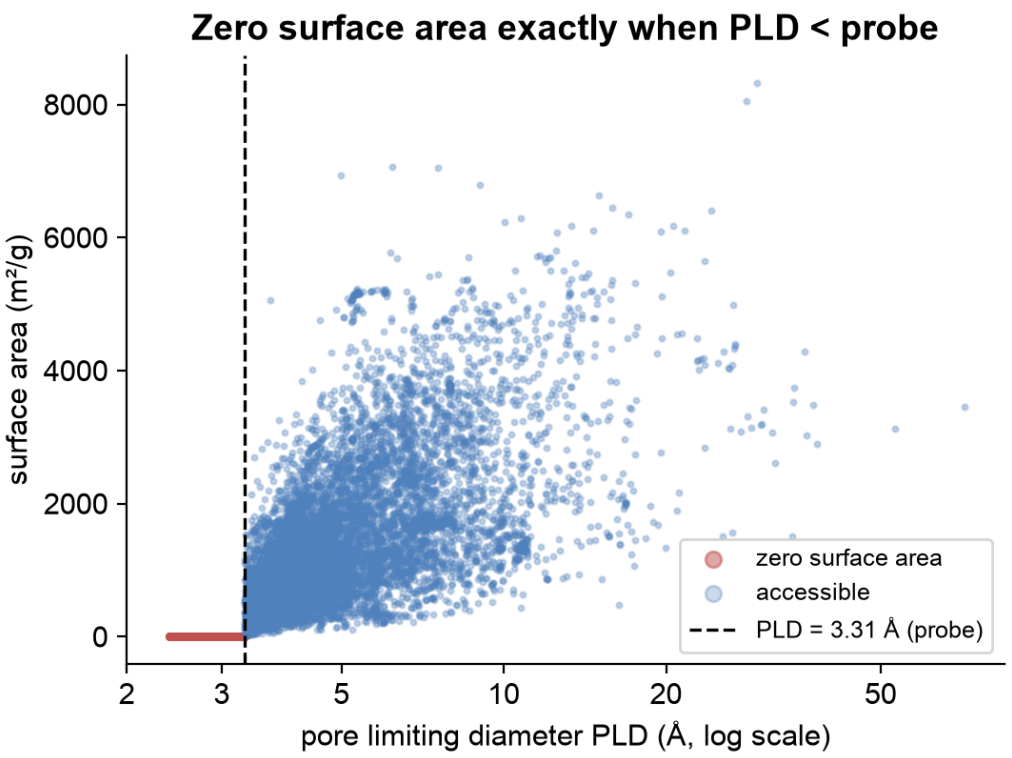
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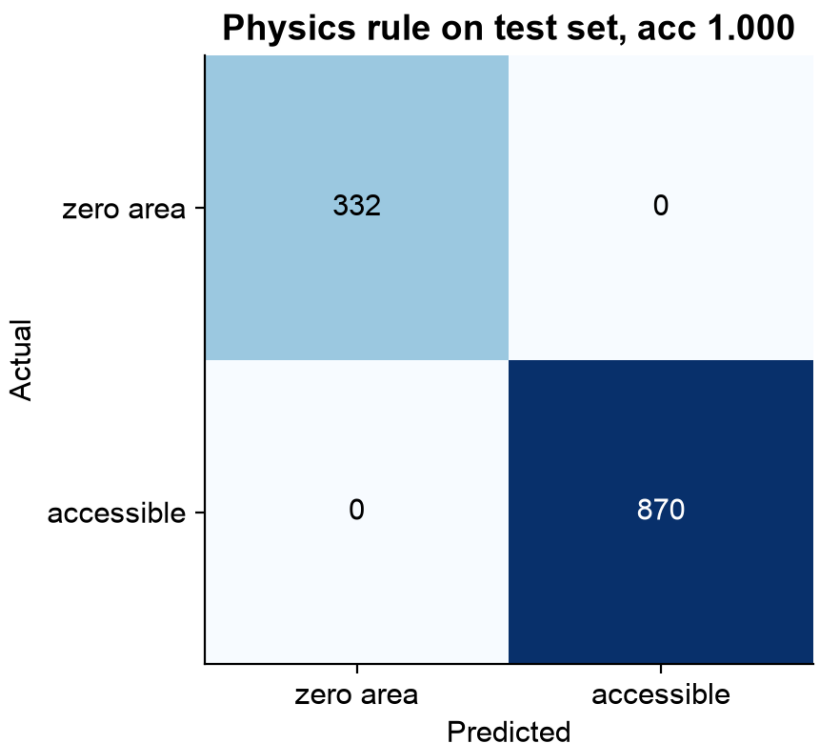
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Thank You

github.com/Sai-Mohith24/MOFChat-AI-Driven-Material-Recommendation-

Appendix: Physics Rule as a Classifier

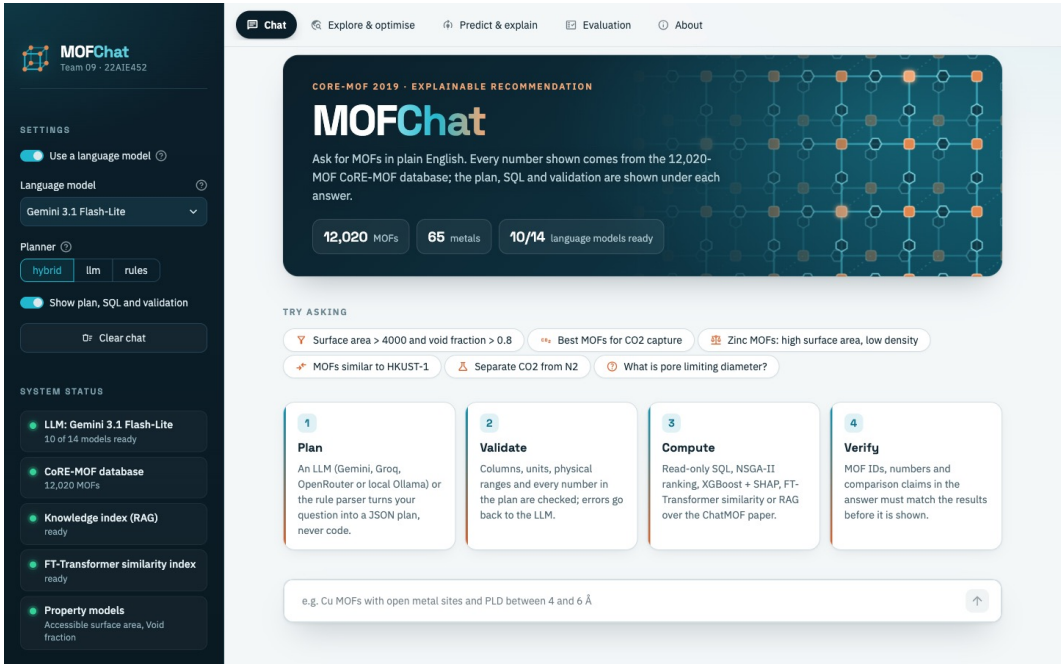




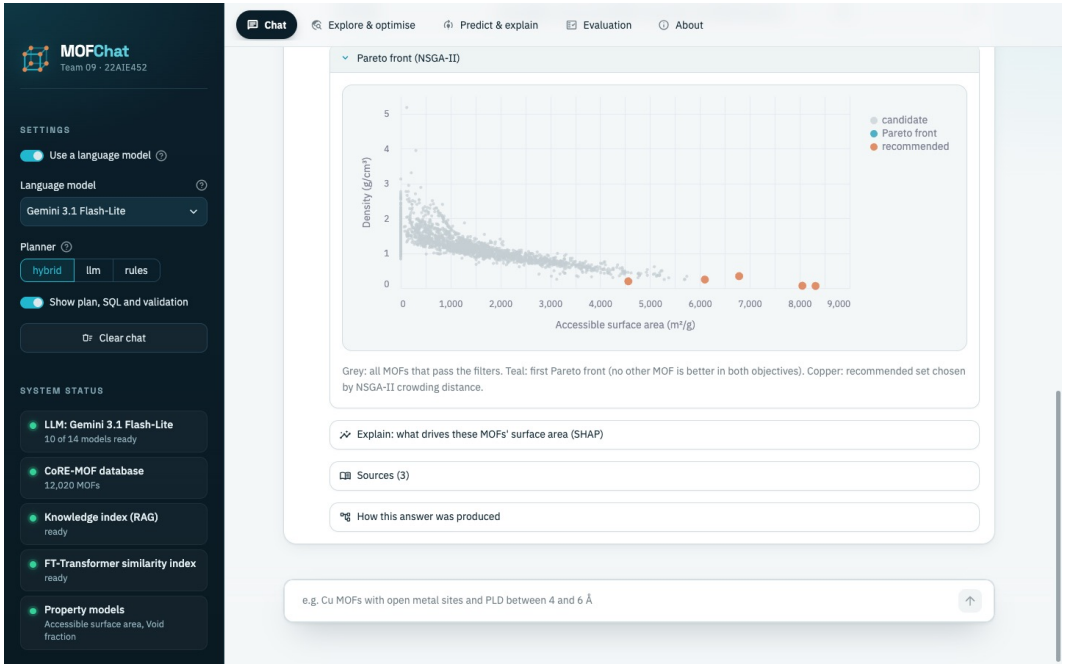
Rule: surface area = 0 if PLD < 3.3107 Å. Test accuracy 100.00 % on 1202 MOFs.

Terms: Probe = the 3.31 Å sphere (size of N2) rolled through the pores to measure them. If the narrowest window (PLD) is smaller than the probe, the probe cannot enter, so the accessible surface area is zero.

Appendix: Application Screenshots



Chat page with model choice and system status



Pareto plot for the zinc MOF question

